4.1 Model Description (Improved Version)

The proposed model implements a simplified Kalman Filter to estimate heading direction using IMU and magnetometer data. For each input window, gyroscope readings are used in the prediction step to update the heading, while magnetometer measurements are used in the correction step to reduce accumulated drift. The model operates sequentially over time and outputs a single heading estimate representing the orientation of the user.

4.2 Layer Structure

Component	Details
State variables	The filter tracks a single state variable, which is the heading angle (θ). This represents the orientation of the user at each time step.
Measurement model	The gyroscope z-axis is used in the prediction step to estimate changes in heading, while the magnetometer (x and y axes) is used in the update step to provide an absolute heading reference.
Filter type	Simplified Kalman Filter (implemented with nonlinear angle wrapping, similar to an EKF for circular data).
Noise parameters	Process noise (Q) and measurement noise (R) are manually tuned hyperparameters. In this implementation, both Q and R are set to 0.01 to balance prediction stability and measurement correction.

4.3 Trainable Parameters

Total trainable parameters: **Not applicable**

The Kalman Filter is a model-based method and does not contain any trainable parameters. Unlike deep learning models, it does not learn from data through optimisation, but instead uses predefined equations along with manually tuned noise parameters (Q and R).

5.1 Loss Function

No explicit loss function is used in this model. The Kalman Filter is a model-based approach and does not involve a training process or optimisation. Instead, it estimates the heading sequentially using prediction and correction steps based on sensor data.

5.2 Training Hyperparameters

Hyperparameter	Value
Optimizer	Not applicable
Learning rate	Not applicable
Batch size	Not applicable
Number of epochs	Not applicable

5.3 Learning Rate Scheduling

None used.

5.4 Early Stopping

Not used.

5.5 Training Difficulties

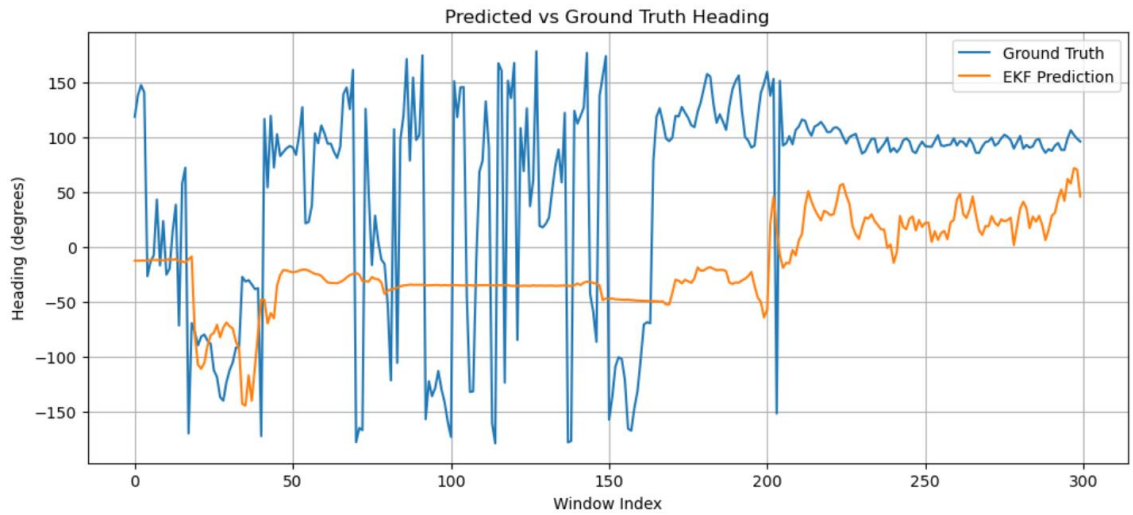
No significant training difficulties were encountered, as the Kalman Filter does not require a training process. However, tuning the process noise (Q) and measurement noise (R) parameters required some experimentation to achieve stable and accurate heading estimates.

6.1 Quantitative Results

Metric	Value
MAE on test set (degrees)	**86.79**
RMSE on test set (degrees)	**102.13**
Approximate training time	Not applicable (no training required)

The results show the overall performance of the Kalman Filter on the test dataset using circular error metrics. The model produces a relatively high error, indicating that while it can provide a basic estimate of heading, its accuracy is limited. This is expected, as the Kalman Filter relies on simple motion assumptions and does not learn complex patterns from data.

6.2 Predicted vs Ground Truth Trajectory



The predicted heading was compared with the ground truth over a portion of the test data. From the plot, it is clear that the Kalman Filter is able to follow the overall direction in some smoother sections, but there are noticeable gaps between the predicted and actual values in many parts.

When the ground truth changes quickly, especially during sharp turns, the prediction does not respond as fast and tends to change more gradually. This makes the predicted curve look smoother, but it also means it cannot capture sudden movements accurately. This behaviour is expected, since the Kalman

Filter tries to balance between the prediction from the gyroscope and the correction from the magnetometer, which naturally introduces some delay.

Overall, the model gives a rough estimate of the heading, but its accuracy is limited, particularly in situations where the movement is more complex or changes rapidly.

6.3 Loss Curve

Not applicable, as the Kalman Filter does not involve a training process.

6.4 MAE Breakdown by Scenario

Scenario	MAE (degrees)
Straight walking	Not computed
Sharp turns	Not computed
Short trajectories	Not computed
Long trajectories	Not computed

A separate breakdown of MAE for different scenarios was not done in this work. The dataset does not directly label different types of movement, such as straight walking or sharp turns, which makes it difficult to evaluate them individually. To do this properly, the data would need to be divided based on motion patterns, which was outside the scope of this implementation.

7.1 Where the Model Performs Well

The model performs reasonably well in segments where the movement is smooth and does not change direction frequently. In these parts of the trajectory, the predicted heading follows the general trend of the ground truth and remains relatively stable. This is mainly because the Kalman Filter combines the gyroscope prediction with magnetometer correction, which helps reduce noise and produces a smoother estimate. As a result, in steady walking conditions, the model is able to maintain a consistent heading without large fluctuations.

7.2 Where the Model Struggles

The model struggles in situations where the heading changes rapidly, such as during sharp turns or irregular movements. From the results, it can be seen that the ground truth often shows sudden changes, while the predicted values respond more slowly and appear smoother. This leads to noticeable errors in those segments. The filter also has difficulty handling noisy or unstable sensor readings, especially from the magnetometer, which can affect the correction step. Overall, the model finds it challenging to accurately track complex motion patterns and fast directional changes.

7.3 Loss Curve Interpretation

Not applicable, as the Kalman Filter does not involve a training process or loss curve.

7.4 Architecture vs Training — Root Cause of Limitations

The limitations observed in this model are mainly due to the architecture itself rather than training choices. Since the Kalman Filter is based on predefined equations and assumptions, it does not learn from data and cannot adapt to complex or nonlinear motion patterns. It assumes relatively simple motion behaviour and relies heavily on the quality of sensor inputs. Because of this, it struggles in scenarios where the motion is highly dynamic or the sensor data is noisy. In contrast, learning-based models can capture more complex relationships, which explains the difference in performance.

7.5 One Change You Would Make

If more time was available, one improvement would be to combine the Kalman Filter with a learning-based model, such as an LSTM or Transformer. This would allow the model to benefit from both approaches — using the Kalman Filter for stable prediction and a neural network to better handle complex patterns and sudden changes in motion. This type of hybrid approach could potentially improve accuracy while still maintaining some level of stability in the predictions.